

Valid/ready handshake

A streaming beat moves when source and sink agree in the same cycle

Starter beat

- Starter preset: both assert on cycle zero
- Valid and ready are one, data is hex A5, and fire is one, one beat transfers immediately
- The wave table shows eight cycles; click valid or ready cells to toggle
- Step through cycles to see when fire lights up
- Try source-stall preset, valid rises late while ready waits
- Try sink-stall, valid held high until ready finally rises

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / Handshake

INTERACTIVE TOOL

Bus handshake animator

Step a conceptual valid/ready transfer: data moves only when both are high in the same cycle. Try source-stall and sink-stall patterns. Starter: both assert together → transfer on cycle 0.

Starter example: valid=ready=1 on cycle 0 with data 0xA5 — one beat transfers. Toggle cells or try stall presets.

Challenge 0 / 22 cleared

Quiz: transfer: In valid/ready handshake, a beat transfers when...

☐ valid and ready are both 1 in the same cycle

☐ only valid is 1

☐ only ready is 1

☐ the clock stops

Show hint

Check

Next

Load challenge setup

Idle

Quiz: transfer

Quiz: valid

Quiz: ready

Quiz: source stall

Quiz: sink stall

Quiz: hold

Quiz: where used

Quiz: combo ready

Transfer @@

One beat

Source stall xfer

Sink stall xfer

Three beats

Zero transfers

Handshake starter

Workbook practice

- On paper, draw valid and ready waveforms for one successful beat and one sink stall
- Write the fire equation in words
- Tabulate four cycles showing valid, ready, and whether a transfer happens
- Explain what the source should do while valid is high and ready is still low
- Name one place you might see this handshake style on chip

Pitfalls to watch

- Do not assume valid alone moves data, both sides must agree
- Do not toggle data while waiting for ready; stability matters for real RTL
- Avoid combinational loops where ready depends on valid in a way that creates a cycle
- This lab is conceptual timing literacy, not a full protocol checker or VIP
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, run together, source-stall, and sink-stall presets and count transfers
- On paper, sketch a two-beat back-to-back burst
- When you are ready, take the short quiz, then continue to the block-diagram integrator